

Sumin Hong

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INTERESTS

Human-AI interaction, Visual generation, Multimodal fusion, Behavior modeling

EDUCATION

University of Notre Dame

Ph.D, Computer Science and Engineering

Advisor: Dr. Toby Jia-Jun Li

Notre Dame, IN

Aug. 2024 – (May. 2029)

Northumbria University

*B.S, Information Technology Management (*Dual Degree)*

Honors: First-class

London, UK

Mar. 2021 – Feb. 2024

Seoul National University of Science and Technology (SeoulTech)

B.S, Information Technology Management

Honors: Summa Cum Laude

Seoul, KR

Mar. 2017 – Feb. 2024

PUBLICATIONS

MORA: AI-Mediated Story-Based practice for Speech Sound Disorder from Clinic to Home

(Submitted) *IUI 26: ACM Conference on Intelligent User Interface*

Sumin Hong, Xavier Briggs, Qingxiao Zheng, Yao Du, Jinjun Xiong, Toby Jia-Jun Li

Vistoria: A Multimodal System to Support Fictional Story Writing through Instrumental Text-Image Co-Editing

(Submitted) *CHI 26: ACM Conference on Human Factors in Computing Systems*

Kexue Fu, Jingfei Huang, Long Ling, Sumin Hong, Yihang Zuo, Ray LC, Toby Jia-Jun Li

CreativeConnect: Supporting Reference Recombination for Graphic Design Ideation with Generative AI

CHI 24: ACM Conference on Human Factors in Computing Systems

DaEun Choi, Sumin Hong, JeongEun Park, John Joon Young Chung, Juho Kim

Class-wise Buffer Management for Incremental Object Detection: Effective Buffer Training Strategy

ICASSP 24: IEEE International Conference on Acoustics, Speech and Signal Processing

Junsu Kim, Sumin Hong, Chanwoo Kim, Jihyeon Kim, Yihalem Yimolal Tiruneh, Jeongwan On, Jihyun Song, Sunhwa Choi, Seungryul Baek

Automatic Examination of Condition of Used Books with YOLO-based Object Detection Framework

Computer Systems Science and Engineering, 47(2), 1611-1632, 2023 (JIF=4.397)

Sumin Hong, Jin-woo Jeong

Ensemble of Multi-task Learning Networks for Facial Expression Recognition In-the-Wild with Learning from Synthetic Data

ECCV 23 4th Workshop on Affective Behavior Analysis in-the-Wild: IEEE/CVF European Conference on Computer Vision

Jae-yeop Jeong¹, Young-gi Hong¹, Sumin Hong¹, Jiyeon Oh¹, Jin-woo Jeong

POSTERS

Proactive Support through floating windows in Information Foraging with User Attention Modeling via Eye-tracking, *CHI 25 Computational UI Workshop*, Sumin Hong, Brandon Foresth, Toby Li

An Effective Buffer Training Strategy for Class-Incremental Object Detection, *CVPR 23 Workshop on Continual Learning in Computer Vision: IEEE/CVF Conference on Computer Vision and Pattern Recognition (Non-archival)*, Junsu Kim, Sumin Hong, Chanwoo Kim, Hoseong Cho, Yihalem Yimolal Tiruneh, Jihyeon Kim, Eunseo Kim, Jihyun Song, Sunhwa Choi, Seungryul Baek

Visualizing Instructor's Gaze Information for Online Video-based Learning: Preliminary Study, *ETRA 22 Workshop on Eye Tracking in Learning and Education: ACM Symposium on Eye Tracking Research and Applications*, Daun Kim¹, Jae-yeop Jeong¹, Sumin Hong, Namsub Kim, Jin-woo Jeong

EXPERIENCE	Lucid Dream	Seoul, KR Nov. 2023 – Feb. 2024
	<i>Applied ML/AI Engineer</i> Work on Posture-Skeleton-based Anomaly Detection Modeling for Rehabilitation of children with Developmental Disabilities	
	KAIST Interaction Lab (KIXLAB)	Daejeon, KR Jun. 2023 – Feb. 2024
	<i>Research Intern</i> Mentors: Juho Kim, DaEun Choi Research on AI-based Creativity Support for Visual Design Ideation & Comic Drawing	
	UNIST Vision and Learning Lab (UVLL)	Ulsan, KR Jan. 2023 – Jun. 2023
	<i>Research Intern</i> Mentors: Seungryul Baek, Junsu Kim Research on Object Detection Modeling & Evaluation with a Class-incremental Feature-retrieving Strategy	
	SeoulTech Interaction Lab (IXLAB)	Seoul, KR Jun. 2021 – Dec. 2022
	<i>Research Intern</i> Mentors: Jin-woo Jeong, Jae-yeop Jeong Research on Affective Computing, Eye-tracking, Defect Detection, Sound source & Saliency viz in 360 video	
HONORS AND AWARDS	Faculty Choice Research Award CSE Poster Competition University of Notre Dame	Mar. 2025
	Graduate Best Award SeoulTech	Dec. 2023
	Full Tuition Scholarship merit-based SeoulTech	2017, 2023
	SeAH Haiam Scholarship given to selected student SeAH Corporate	2023
	3rd place 4th Workshop on Affective Behavior Analysis in-the-wild ECCV 2022	Jul. 2022
	Outstanding Paper Award Winter Annual Conference of KICS	Jan. 2022
TEACHING	Teaching Assistant CSE20110: Discrete Mathematics University of Notre Dame	Fall 2024
	Teaching Assistant ITM146091: Programming Language SeoulTech	Fall 2020
ACADEMIC SERVICE	Reviewer CHI 26, CHI LBW 25	